Facilities Management in Healthcare

Facilities management covers hospital infrastructure systems such as power, water, medical gases, HVAC, safety systems, regulations, and emergency preparedness to ensure patient safety and continuous care.

Why Facilities Management Matters in Hospitals?

- Hospital facilities directly affect **patient safety and outcomes**
- Poor facility management can cause:
  - Infections
  - Treatment interruptions
  - Equipment failure
- Some infections can lead to **serious illness or death**

Environmental Pathogens in Healthcare Facilities

- Poorly maintained systems allow environmental pathogens to grow and spread
- Examples:
  - Aspergillus → grows in damp construction areas (airborne)
  - Legionella → grows in water systems (waterborne)



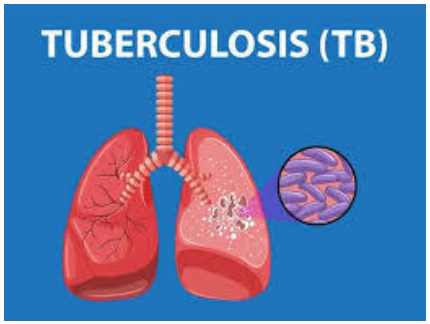
- These can cause severe infections in vulnerable patients
- Unlike common healthcare-associated bacteria, these pathogens originate from air and water systems



Airborne Infection Risks

- Some diseases spread through air if ventilation is poor:
  - Tuberculosis (TB)
  - Varicella zoster virus (chickenpox)





- Proper HVAC systems are essential to:
  1. Control airflow
  2. Remove contaminated air
  3. Protect patients and staff

Importance of Following Standards and Guidelines

- Failure to follow standards may lead to:
  - Unsafe dialysis water
  - Poor operating room ventilation
  - Improper cleaning and disinfection
- Results:
  - Increased infection rates
  - Poor patient outcomes
  - CDC provides healthcare facility safety guidance

Who Sets the Rules for Hospitals?

- Hospitals must follow many regulations from:
  - Joint Commission
  - CMS (Centers for Medicare & Medicaid Services)
  - NFPA (National Fire Protection Association)
  - *Local and state health authorities*
- These organizations affect:
  - Design
  - Construction
  - Operation
  - Safety procedures

Important NFPA Codes in Healthcare Facilities

- Most commonly used codes:
  - **NFPA 70** → National Electrical Code
  - **NFPA 99** → Health Care Facilities Code
  - **NFPA 110** → Emergency Power Systems
  - **NFPA 111** → Stored Energy Emergency Power
- These codes regulate:
  - Electrical safety
  - Medical gas systems
  - Emergency power



Examples of Safety Requirements from Codes

- Codes require:
  - Ground fault protection in wet areas
  - Automatic power transfer switches
  - Safe medical gas installation
  - Proper alarm systems
  - Backup electrical supplies
- Goal: **continuous and safe patient care**

Emergency Power and Generators

- Hospitals must have backup power systems
- Different **generator** types:
  - Diesel
  - Gas
  - Battery systems
- Each has:
  - Specific fuel requirements
  - Safety procedures
- Backup systems must start **automatically**

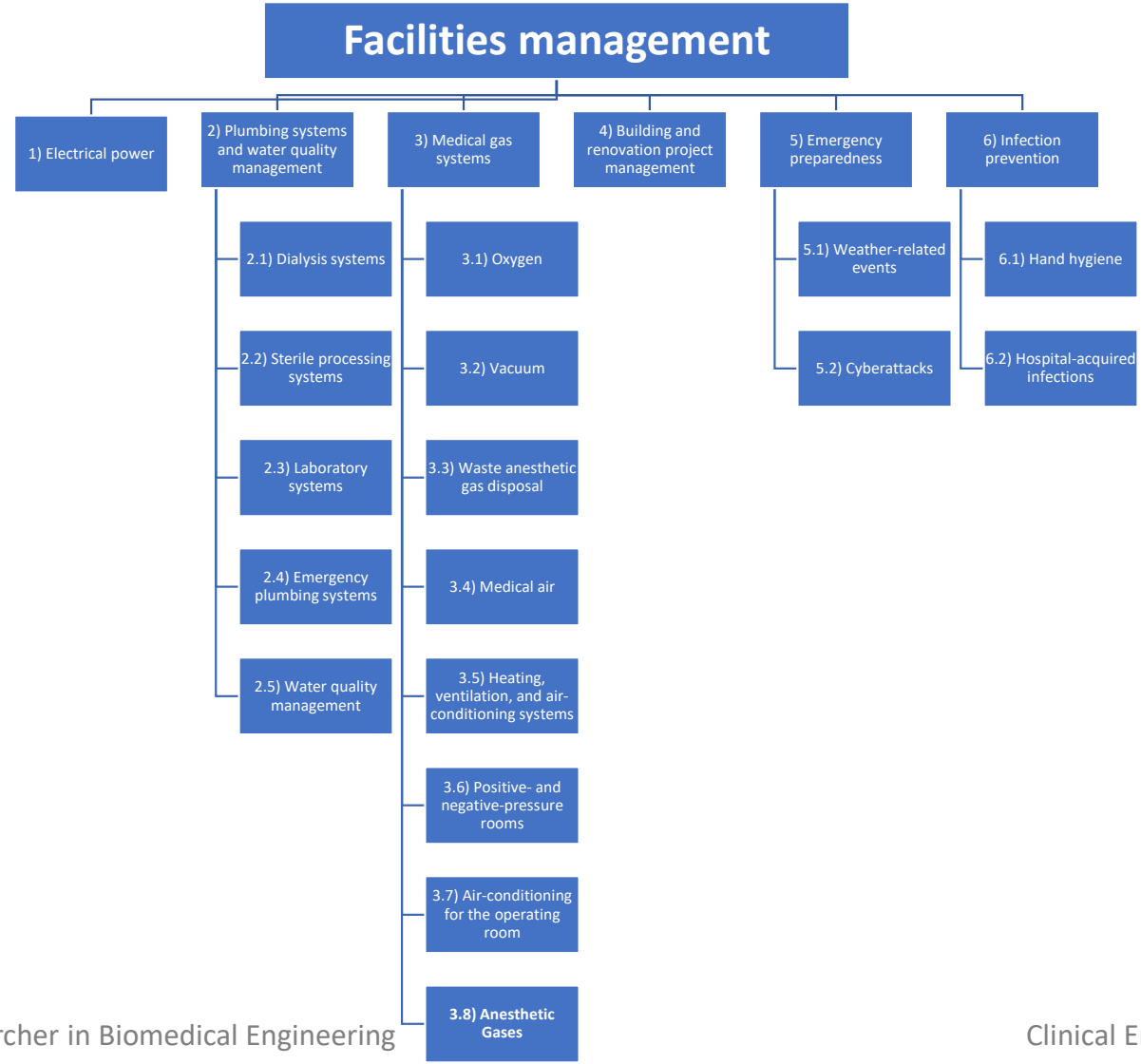


Maintenance and Testing Requirements

- Regulations require:
  - Preventive maintenance schedules
- Regular testing of:
  - Generators
  - Transfer switches
  - Medical gas systems
  - Heating, ventilation, and air conditioning (HVAC) systems
- Purpose:
  - Detect problems early
  - Prevent system failure during emergencies

Role of Clinical & Facilities Engineers

- Engineers are responsible for:
  - Monitoring system performance
  - Ensuring compliance with standards
  - Coordinating inspections and testing
  - Managing risks to patients and staff
  - Facilities management is part of **patient safety strategy**


Facilities management
1) Electrical power
2) Plumbing systems and water quality management
3) Medical gas systems
4) Building and renovation project management
5) Emergency preparedness
6) Infection prevention
2.1) Dialysis systems
2.2) Sterile processing systems
2.3) Laboratory systems
2.4) Emergency plumbing systems
2.5) Water quality management
3.1) Oxygen
3.2) Vacuum
3.3) Waste anesthetic gas disposal
3.4) Medical air
3.5) Heating, ventilation, and air-conditioning systems
3.6) Positive- and negative-pressure rooms
3.7) Air-conditioning for the operating room
3.8) Anesthetic Gases
5.1) Weather-related events
5.2) Cyberattacks
6.1) Hand hygiene
6.2) Hospital-acquired infections



1 - Electrical power

1 - Electrical power - Why Electrical Power Is Critical

- Hospitals need **continuous, interruption-free power**
- Most medical equipment depends on electricity:
  - Imaging
  - Life support
  - Monitoring systems
- Any power failure can immediately threaten **patient safety**

1 - Electrical power - Why Main Power Is Not Enough?

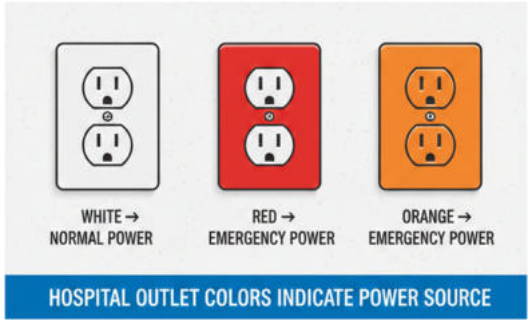
- Hospitals normally use the **public electrical grid**
- Grid power can fail due to:
  - Weather events
  - System faults
  - Brownouts
- Therefore, hospitals **must have backup power systems**

1 - Electrical power - Backup Power: What It Really Supports

- Backup power does **not supply the whole hospital**
- It is designed only for:
  - Life safety systems
  - Critical patient care areas
- Engineers must decide **which loads are essential**

1 - Electrical power - Power Outlet Color Coding (Engineering Practice)

- Hospitals use outlet colors to indicate power source:
  - White → normal (main) power
  - Red / Orange → emergency power
- Only devices on emergency outlets continue working after outage
- Clinical engineers must ensure:
  - Critical equipment is plugged into correct outlets



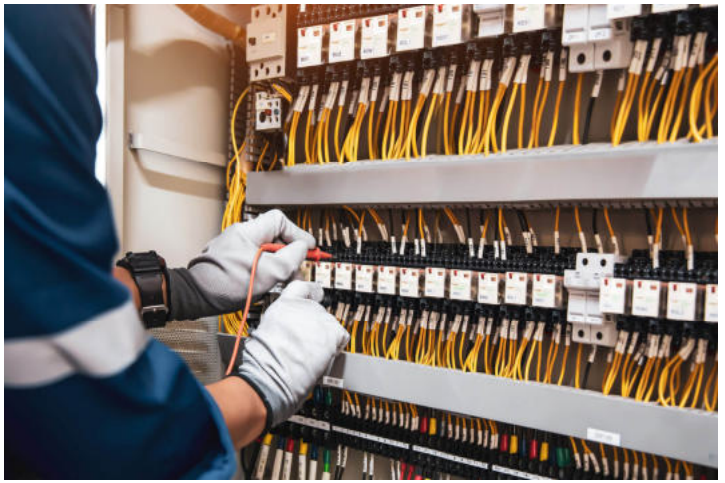


1 - Electrical power - Emergency Power System Structure

- Emergency power is divided into two main branches:
- **1) Life Safety Branch**
- Supports:
  - Emergency lighting
  - Exit signs
  - Elevators
  - Automatic doors
- **2) Critical Branch**
- Supports:
  - Patient care equipment
  - Nurse call systems
  - Blood banks
  - Special medical power systems

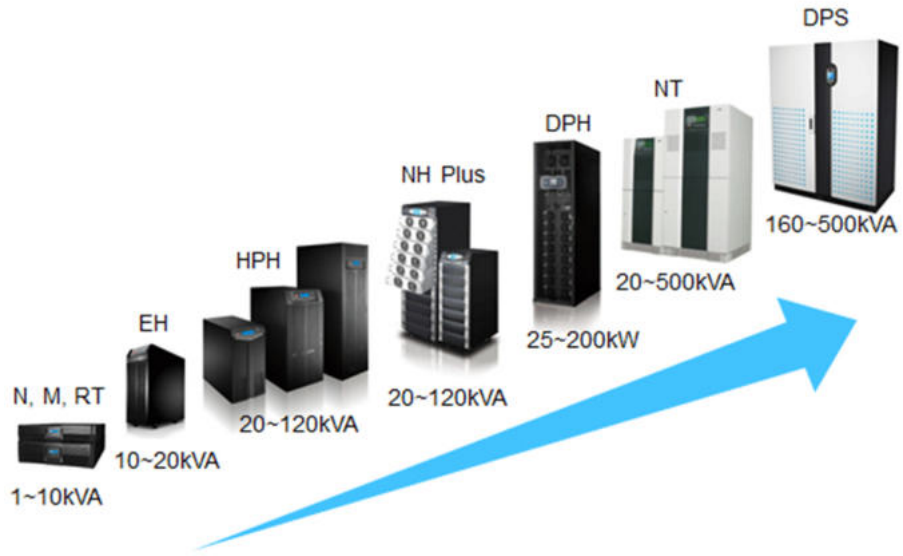
1 - Electrical power - Electrical Design Considerations

- Engineers must consider:
  - Total electrical load during outage
  - Generator capacity
- Power transfer time:
  - From a few seconds to minutes
- Risk of overload during switching



1 - Electrical power - Uninterruptible Power Supplies (UPS)

- Some devices cannot tolerate even short power loss
- UPS systems are used to:
  - Smooth power transition
  - Prevent device shutdown or data loss
- Common for:
  - IT systems
  - Critical monitoring equipment



2 - Plumbing systems and water quality management

2 - Plumbing and water - Why Hospital Plumbing Is Different ?

- **Why Hospital Plumbing Is Different**
- Hospital plumbing systems are more complex than normal buildings
- Must support:
  - Patient care
  - Medical procedures
  - Infection control
- Systems require **special design and maintenance**

2 - Plumbing and water - Medical Equipment Has Special Water Needs

- Different hospital departments require:
  - Specific water quality
  - Controlled temperature
  - Reliable supply
- Maintenance is more critical than in:
  - Offices
  - Hotels
  - Residential buildings

2 - Plumbing and water - Water Temperature and Patient Safety

- Typical buildings:
- Hot water: **49–60°C** (120–140°F)
- Hospitals:
- Handwashing & showers: **max 43°C** (110°F)
- Psychiatric & dementia units: **max 38°C** (100°F)
- Purpose:
- Prevent burns
- Protect vulnerable patients

2 - Plumbing and water - Engineering Controls for Temperature Limits

- To maintain safe temperatures, hospitals use:
- Mixing valves
- Temperature regulators
- Continuous monitoring
- Engineering goal:
- Balance between:
  - Infection control
  - Scald prevention

2 - Plumbing and water - Department-Specific Plumbing Requirements



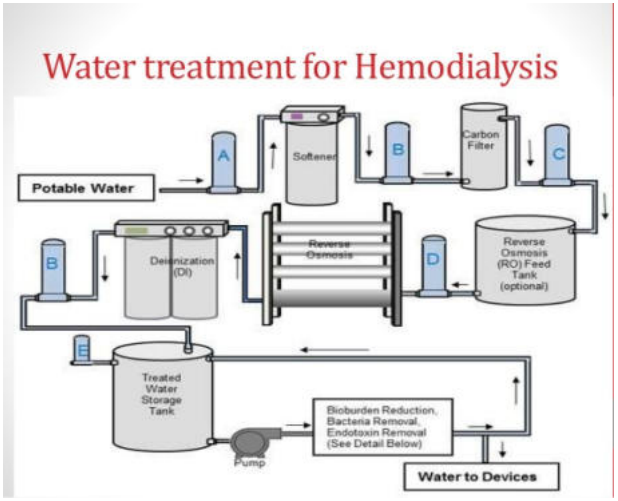
- Different departments need special plumbing systems:
  - Dialysis
  - Sterile processing
  - Laboratories
  - Emergency systems
- Each has:
  - Different risks
  - Different quality standards
- These will be discussed in the next subtopics



2.1 - Dialysis Systems

2.1 - Dialysis Systems -Why Water Quality Is Critical in Dialysis

- Dialysis patients are exposed to **large volumes of water**
- Any contamination can directly enter the bloodstream
- Therefore, dialysis water must be **much purer than drinking water**



2.1 - Dialysis Systems - How Hospitals Achieve Ultrapure Water

- Two main system options:
  - **Central Reverse Osmosis (RO) System**
  - **Portable Filtration / RO Systems**
- Both aim to:
  - Remove bacteria
  - Remove chemicals
  - Remove particles and minerals





2.1 - Dialysis Systems - Central RO vs Portable Systems

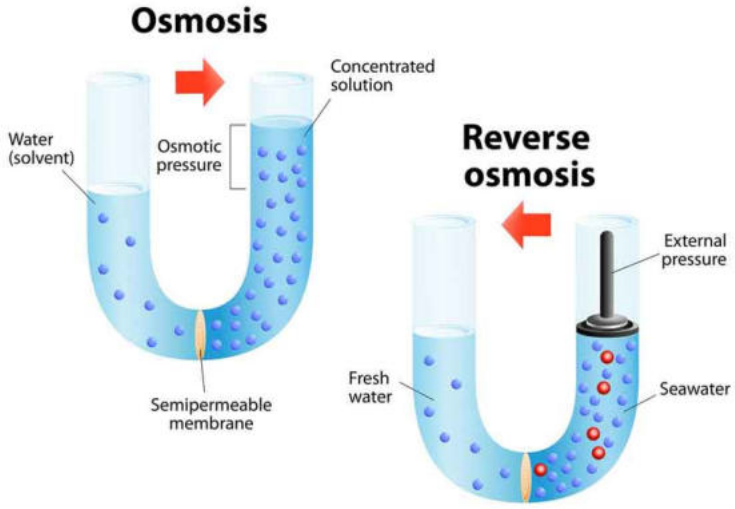
- **Central RO Systems**
  - Used in large dialysis centers
  - High installation cost
  - High maintenance needs
  - Serve many dialysis machines
- **Portable RO Systems**
  - Smaller units
  - Connect to tap water
- Used in:
  - Homes
  - Small clinics
  - Hospitals for temporary dialysis





2.1 - Dialysis Systems -Reverse Osmosis (RO) System: How It Works

- RO systems use:
  - High-pressure pump
  - Semi-permeable membrane
- Water passes through membrane → Contaminants are rejected and flushed away
- Additional components may include:
  - Sediment filters (large particles)
  - Carbon filters (chemicals)
  - Water softeners (calcium & magnesium)



2.1 - Dialysis Systems - What Happens to Waste Water?

- Contaminants are removed as **reject (waste) water**
- Waste water is discharged to:
  - Drain system
- Engineers must consider:
  - Drain capacity
  - Infection control
  - Environmental regulations

2.1 - Dialysis Systems - Maintenance and Monitoring Requirements

- Both systems require:
- Regular filter replacement
- Membrane testing
- Microbial testing of water
- Continuous quality monitoring
- Failure in maintenance = **direct patient risk**

2.2 - Sterile Processing Systems

2.2 – Sterilization - What Is Sterile Processing?



- Responsible for:
  - Cleaning
  - Disinfecting
  - Sterilizing surgical instruments
- Directly affects:
  - Surgical safety
  - Infection prevention
- This is a **high-risk, high-dependency department**

2.2 – Sterilization - High Demand on Water and Steam

- Sterile processing uses:
  - Large volumes of water
  - Large amounts of steam
- Plumbing and drainage systems must be designed to:
  - Handle heavy loads
  - Prevent backups and failures
- Engineering design must consider **continuous operation**



2.2 – Sterilization - Temperature Is Critical for Sterilization

- Sterilizers require hot water and steam at:
  - **71–82°C** (160–180°F)
- Minimum temperature must be:
  - Maintained during the full cycle
- If temperature drops:
  - Sterilization may fail
  - Instruments become unsafe
- Sterilizers may be installed in:
  - Sterile Processing Department (SPD)





2.3 - Laboratory Systems

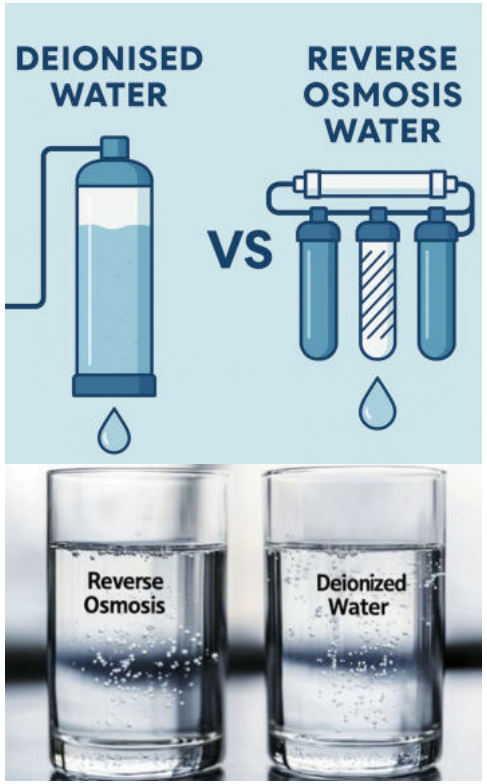
2.3 - Laboratory Systems - Why Labs Need Special Water Systems

- Hospital labs perform:
  - Pathology tests
  - Chemical analysis
  - Diagnostic measurements
- Many tests require **high-purity water**
- Normal tap water is **not sufficient**



2.3 - Laboratory Systems - Types of Purified Water Used in Labs

- Depending on equipment and tests, labs may require:
  - RO water (Reverse Osmosis)
  - Deionized (DI) water
- Used for:
  - Analyzers
  - Reagent preparation
  - Equipment cleaning
- Water quality directly affects **test accuracy**



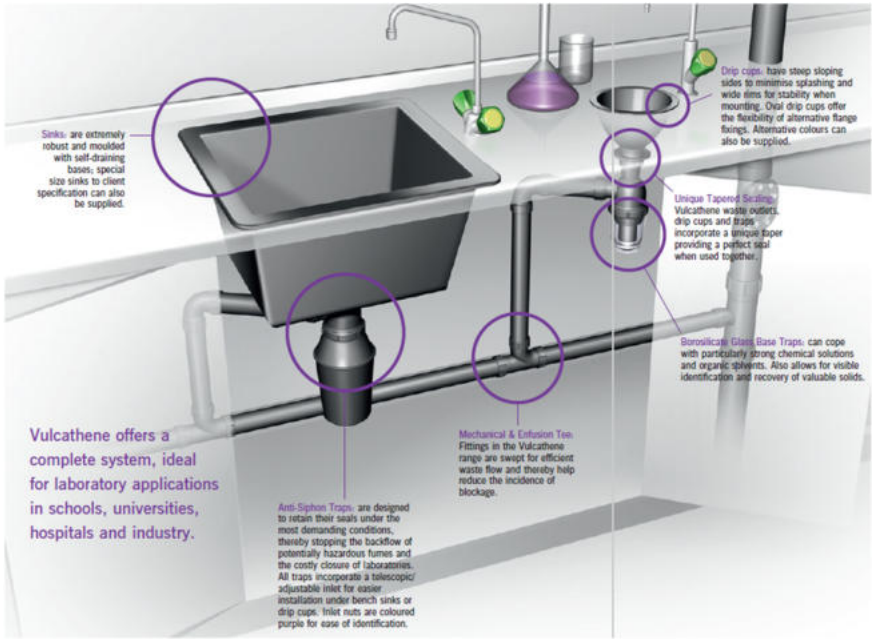
2.3 - Laboratory Systems - Wastewater Is Also a Safety Issue

- Laboratory wastewater may contain:
  - Chemicals
  - Biological materials
  - Foaming agents
- Drainage systems must be designed to:
  - Prevent contamination
  - Protect public sewer systems



2.3 - Laboratory Systems - Waste Treatment at Point of Use

- Some labs require extra systems at sinks or drains:
  - Acid-neutralizing systems
  - Antifoam devices
  - Special drain covers
- These systems:
  - Protect plumbing
  - Reduce environmental hazards





2.4 - Emergency Plumbing Systems

2.4 - Emergency Plumbing - Why Emergency Plumbing Is Needed



- Some hospital areas use:
  - Chemicals
  - Corrosive materials
  - Biological agents
- Accidental exposure can injure:
  - Eyes
  - Skin
- Immediate washing can **prevent serious damage**





2.4 - Emergency Plumbing - Where Are Emergency Systems Required?

- Emergency eyewash and showers are needed in areas like:
  - Laboratories
  - Pharmacy preparation rooms
  - Sterile processing
  - Maintenance and cleaning areas
- Requirement depends on:
  - Type of work
  - Chemicals used (from Safety Data Sheets)

2.4 - Emergency Plumbing -OSHA Legal Requirements

- Occupational Safety and Health Administration (OSHA) requires:
  - Quick access to flushing facilities
  - When eyes or body may contact harmful materials
- Key point:
  - Emergency washing must be **immediately available in the work area**
  - This is a **legal safety requirement**, not optional

2.4 - Emergency Plumbing -Why ANSI Standards Are Important

- OSHA does not give technical details, so:
- **ANSI Z358.1 standard** defines how systems must work

ANSI specifies:

- Water temperature
- Flow rate
- Spray pattern
- Hands-free operation
- Engineers must follow **ANSI for design and installation**







2.4 - Emergency Plumbing - Location and Access Requirements

- Emergency equipment must be:
- Reachable within **10 seconds**
- About **17 meters (55 feet)** from hazard
- Path must be:
  - Straight
  - Free of obstacles
- Design goal: **no delay during accidents**

2.4 - Emergency Plumbing - Water Temperature Requirements

- Water must be **tepid** to avoid injury and allow long flushing:
- Acceptable range: **16–38°C** (60–100°F)
- Why this matters:
  - Too cold → patient cannot stay long
  - Too hot → can worsen chemical injury

2.4 - Emergency Plumbing - low Rate Requirements

- Safety showers must provide:
  - At least **75 liters per minute** (20 gallons/min)
  - For a full **15 minutes**
- Eyewash systems must:
  - Gently rinse eyes and face
  - Without damaging tissues
  - Engineering design must support **continuous high flow**

2.4 - Emergency Plumbing - Types of Emergency Devices

- Systems may include:
  - Eyewash stations
  - Eye/face wash units
  - Full-body safety showers
  - Combination units
- Most must be:
  - Hands-free once activated
  - Stay on until manually turned off







2.5 - Water Quality Management

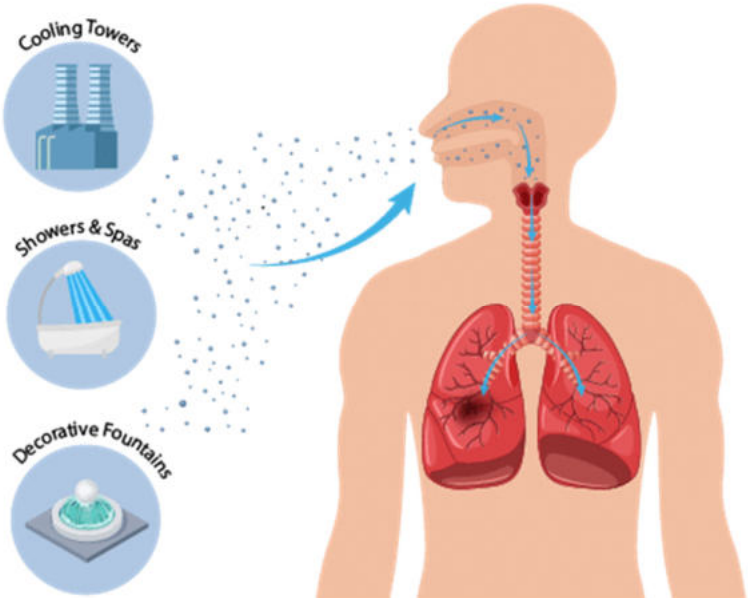
2.5 - Water Quality - Why Water Quality Matters in Hospitals

- Facilities departments manage:
  - Water supply
  - Drainage systems
  - Internal water quality
- Poor water quality can cause:
  - Serious infections
  - Outbreaks in vulnerable patients

2.5 - Water Quality - Main Infection Risk: Legionella

- Legionella bacteria can spread through:
  - Water droplets
  - Aerosols
- High-risk areas:
  - Surgery units
  - Hydrotherapy areas
- High-risk patients:
  - Oncology patients
  - Bone marrow transplant patients





2.5 - Water Quality - Goal of Water Quality Management

- Main goal:
  - Prevent growth and spread of bacteria in plumbing systems
- This protects:
  - Patients
  - Staff
  - Hospital reputation
- Water safety is part of **infection control**



2.5 - Water Quality - Core Principles of Safe Water Systems

- Effective water management includes:
  - Keeping water temperature outside Legionella growth range
  - Preventing stagnant (unused) water
  - Maintaining proper disinfectant levels
  - Keeping plumbing systems clean and functional
  - These are **engineering control strategies**

2.5 - Water Quality - Establish a Water Management Team

- **Building a coordinated, multidisciplinary response**
- **Team Members**
  - Facilities / Engineering
  - Infection Control
  - Safety & Risk Officers
  - Administration / Management
- **Why This Matters**
  - Shared accountability across departments
  - Faster identification and response to water-related risks
  - Stronger coordination for prevention and compliance



3 - Medical gas systems

3 - Medical gas systems - What Are Medical Gas Systems?

- Supply gases needed for:
  - Breathing
  - Anesthesia
  - Patient treatment
- Failure can be **life-threatening**



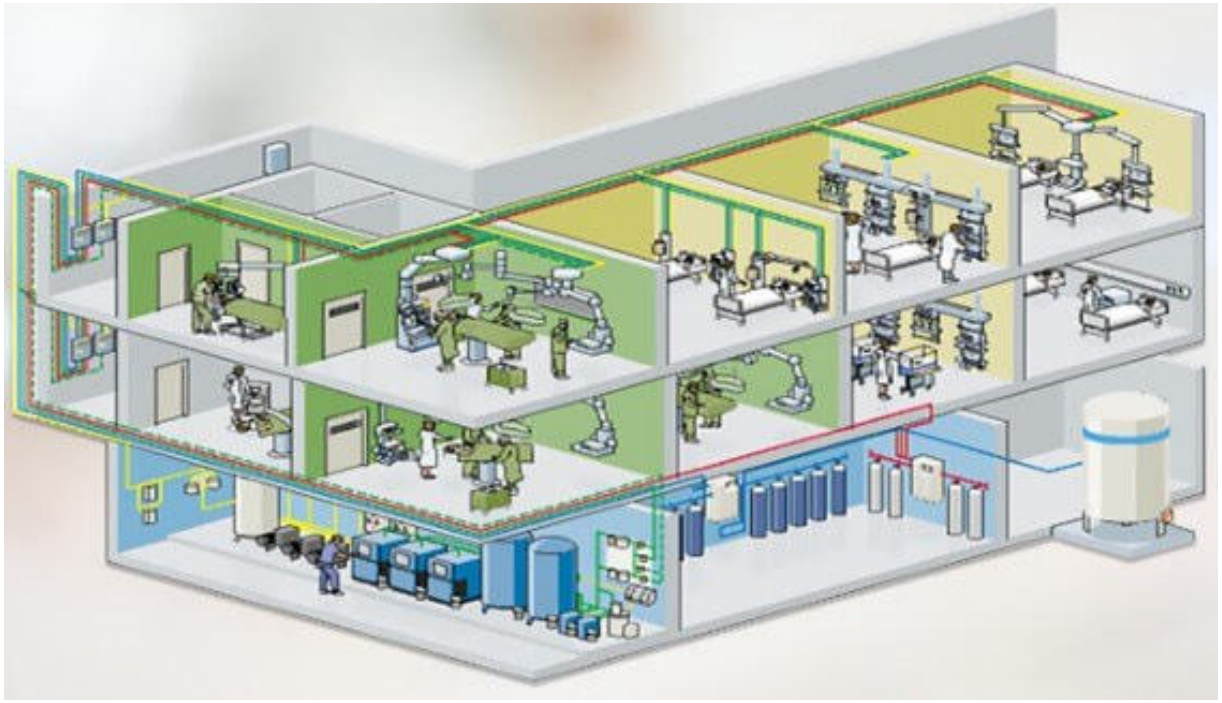
3 - Medical gas systems - Main System Components

- Storage tanks or cylinders
- Manifolds and compressors
- Pumps and piping network
- Alarms and monitoring systems



3 - Medical gas systems - Centralized and Secured Design

- Gases stored in:
  - Special rooms or outside buildings
- Distributed through hospital piping
- Designed for:
  - Continuous supply
  - Emergency backup



Gas	U.S. Color Code	ISO Color Code
Carbon Dioxide	Grey	Grey
He-O_2	Brown & Green	Brown & White
Instrument Air	Red (USA Only)	
Medical Air	Yellow	Black & White
Nitrogen	Black	Black
Nitrous Oxide	Blue	Blue
O_2-He	Green & Brown	White & Brown
Oxygen	Green	White
Vacuum (Suction)	White	Yellow
WAGD (Evac)	Purple	Purple

Material Property	Letter Color on Field Color	Example
Single Gases		
Oxygen USP†	White on Green	OXYGEN 50-55 PSI
Carbon Dioxide†	White on Gray	CARBON DIOXIDE
Nitrous Oxide†	White on Blue	NITROUS OXIDE
Cyclopropane‡	Black on Orange	CYCLOPROPANE
Helium USP†	White on Brown	HELIUM 50-55 PSI
Nitrogen NF†	White on Black	NITROGEN 160-200 PSI
Medical Air USP†	Black on Yellow	MEDICAL AIR
Instrument Air†	White on Red	INSTRUMENT AIR
Waste Anesthetic Gas Disposal†	White on Purple	WASTE ANESTHETIC
Laboratory Air†	Black on White/Yellow Checkerboard	LABORATORY AIR
Laboratory Vaccum†	Black on Black/White Checkerboard[2]	LABORATORY VACUUM
Medical-Surgical Vaccum†	Black on White	MEDICAL VACUUM

Material Property	Letter Color on Field Color	Example
Gas Mixtures of Oxygen USP and Nitrogen NF		
19.5% to 23.5% Oxygen‡	Black on Yellow	20% OXYGEN 80% NITROGEN
All other Oxygen Concentrates†‡	Green on Black	40% OXYGEN 60% NITROGEN
Other Gas Mixtures		
Oxygen USP and Nitrous Oxide USP[1‡]: 52.5% Oxygen USP, 47.5% Nitrous Oxide USP	Defined by User	52.5% OXYGEN 47.5% NITROUS OXIDE
Oxygen USP and Carbon Dioxide USP[1‡]: ≥ 93% Oxygen USP, remainder Carbon Dioxide USP	Gray on Green	93% OXYGEN 7% CARBON DIOXIDE
Oxygen USP and Helium USP[1‡]: ≥ 20% Oxygen USP, remainder Helium USP	Green on Brown	26% OXYGEN 74% HELIUM

† Complies to NFPA 99 color marking standards.
‡ Complies to CGA C-9 color marking standards.
1 Use the background colors of the Single Gases. For other gas mixture pipe labels, use the major gas for label color and use the minor gas color for text color.
2 Use a white box with a black outline behind the text.



3.1 - Oxygen

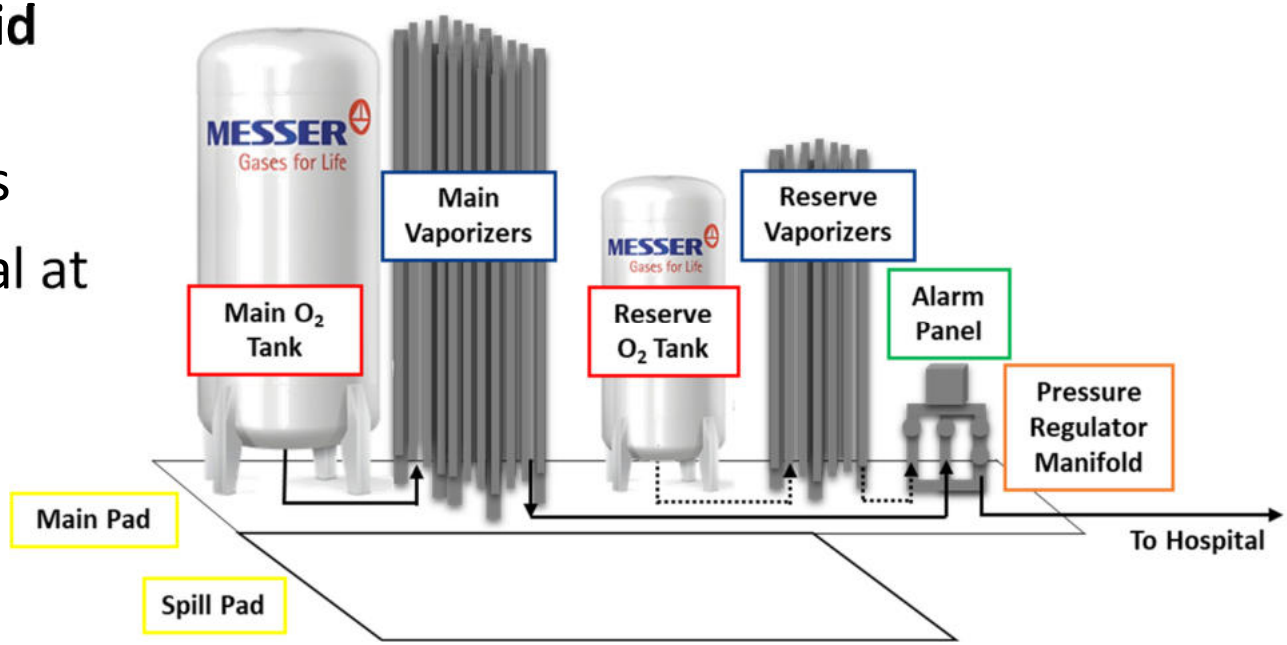
3.1 – Oxygen (O_2) - Why Oxygen Systems Are Critical

- Oxygen is used for:
  - Breathing support
  - Surgery
  - Emergency care
- System failure can be **immediately life-threatening**
- Must be **highly reliable and monitored**



3.1 – Oxygen - Source of Hospital Oxygen

- Oxygen is stored as **liquid oxygen**
- It boils and becomes gas
- Gas is piped into hospital at **controlled pressure**
- Pressure and flow are continuously regulated



3.1 – Oxygen - Main and Backup Supply Requirement

- Hospitals must have:
  - Main oxygen supply system
  - Backup system
- At least **1-day reserve supply**
- At least **1-day emergency backup**
- Purpose:
  - Maintain oxygen even if main system fails

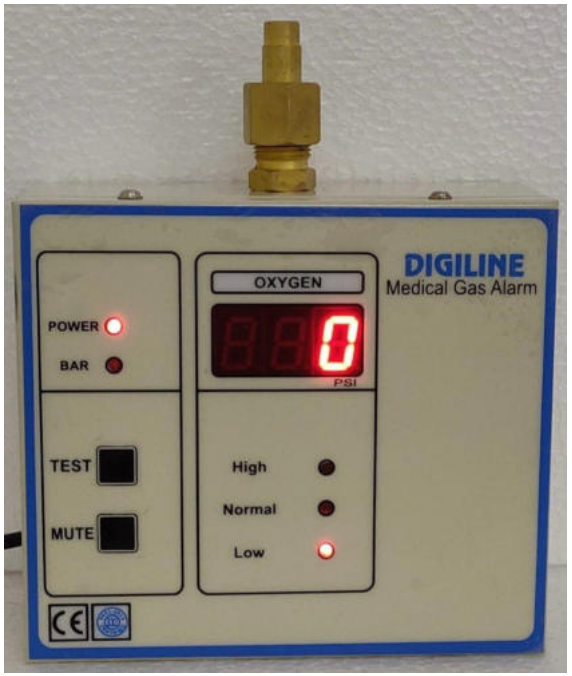


3.1 – Oxygen - Monitoring and Alarm Systems

- System includes:
  - Pressure sensors
  - Volume monitoring
  - Alarm systems
- Alarms are connected to:
  - Building automation system
  - Facilities and security teams
- Any pressure drop = **immediate alert**







3.1 – Oxygen - Distribution Inside the Hospital

- Oxygen flows through:
- Special **medical-grade copper pipes**
- Designed to prevent:
  - Contamination
  - Leaks
- Piping delivers oxygen to:
  - Patient rooms
  - ICUs
  - Operating rooms



3.1 – Oxygen - Zone Valves for Safety and Maintenance

- Piping is divided into **zones**
- Each zone serves:
  - A group of rooms or a floor
- Zone valves allow:
  - Isolation during repairs
  - Maintenance without shutting whole hospital
  - Only authorized staff can access them



3.1 – Oxygen - Pressure Monitoring at Zone Level

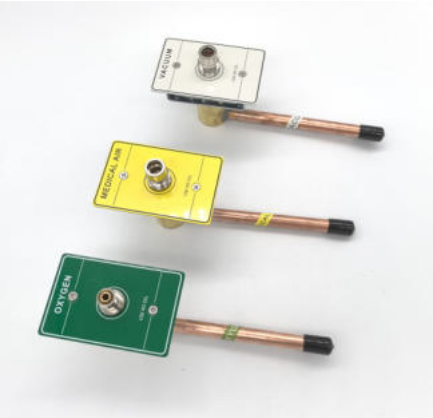
- Zone valves include:
  - Visual pressure gauges
- Engineers can quickly check:
  - Whether pressure is normal
  - If a leak may exist in that zone
  - Important for fast troubleshooting



3.1 – Oxygen - Wall Outlets and Safety Connections

- Oxygen reaches patients through:
  - Wall outlets
- Features:
  - Special quick-connect fittings
  - Color-coded **green** for oxygen
  - Prevents wrong gas connections
  - Patient safety depends on correct fittings





3.1 – Oxygen - Flow Control to the Patient

- At wall outlet, staff attach:
  - Flow or pressure regulator
- Purpose:
  - Control oxygen amount
  - Match patient needs
- Engineering must ensure:
  - Regulators and outlets are compatible and safe





3.2 - Vacuum

3.2 – Vacuum - Purpose of Hospital Vacuum Systems

- Used to remove:
  - Fluids
  - Secretions
  - Gases
- Essential in:
  - Operating rooms
  - ICUs
  - Emergency care
- Vacuum is critical for many procedures

3.2 – Vacuum - Outlets and Safety Connections



- Vacuum outlets:
- Use quick-connect fittings
- Have **unique hose connections**
- Prevent cross-connection with other gases
- Safety design prevents wrong gas hookups



3.3 - Waste Anesthetic Gas Disposal (WAGD)

3.3 – WAGD - Why WAGD Systems Are Needed

- Anesthetic gases are used during surgery
- Not all gas is absorbed by the patient
- Exhaled gas can remain in the room
- Long-term exposure is harmful to staff
- Goal: **protect healthcare workers**



3.3 – WAGD - What the WAGD System Does

- Captures exhaled anesthetic gases
- Removes them from:
  - Patient area
  - Operating room air
- Pumps gases safely outside the building
- This is a staff safety system, not patient suction



3.3 – WAGD - Safety Connections and Color Coding



- WAGD outlets:
- Color-coded purple
- Use unique fittings
- Cannot connect to:
  - Oxygen
  - Medical air
  - Vacuum
- Purpose: prevent dangerous cross-connections

3.3 – WAGD - Gas Discharge Location

- Removed gases are vented:
  - Through roof exhaust
- Important design rule:
- Exhaust must be far from:
  - Air intakes
  - Ventilation inlets
- Prevents gases from re-entering hospital



3.3 – WAGD - Fire and Explosion Risk

- Anesthetic gas mixed with oxygen:
  - Is flammable
- Therefore:
  - Gas must be removed quickly
  - System must maintain sufficient vacuum pressure
- Safety depends on:
  - Proper flow rates
  - Reliable exhaust



3.4 - Medical Air

3.4 - Medical Air - Purpose of Medical Air

- Provides breathing air to patients
- Must have:
  - Correct humidity
  - Clean composition
- Used in:
  - Respiratory therapy
  - Ventilation systems
- This is a **patient life-support utility**



3.4 - Medical Air - Quality and Safety Standards

- Medical air must meet:
  - **USP (United States Pharmacopeia) standards**
- Meaning:
  - No harmful gases
  - No particles
  - Controlled moisture
  - Air quality directly affects patient safety

3.4 - Medical Air - How Medical Air Is Produced

- Process:
  - Air drawn from outside
  - Compressed by medical air compressors
  - Stored in receiver tank
  - Distributed to hospital piping
- Central production allows:
- Continuous monitoring
- Quality control



3.4 - Medical Air - Moisture and Gas Monitoring

- Before reaching patients, air is:
  - Dried to proper dew point
  - Tested for carbon monoxide (CO)
- Why important:
  - Water damages equipment
  - CO is toxic



3.4 - Medical Air - Final Filtration Before Use

- Before reaching outlets:
  - Air passes through final filters
  - Removes remaining particles
- Ensures:
  - Clean, breathable air at bedside



3.4 - Medical Air - Redundancy Is Mandatory

- System must have:
  - Backup compressors
  - Backup dryers
- Reason:
  - Medical air must be available **24/7**
  - Single-point failure is not acceptable

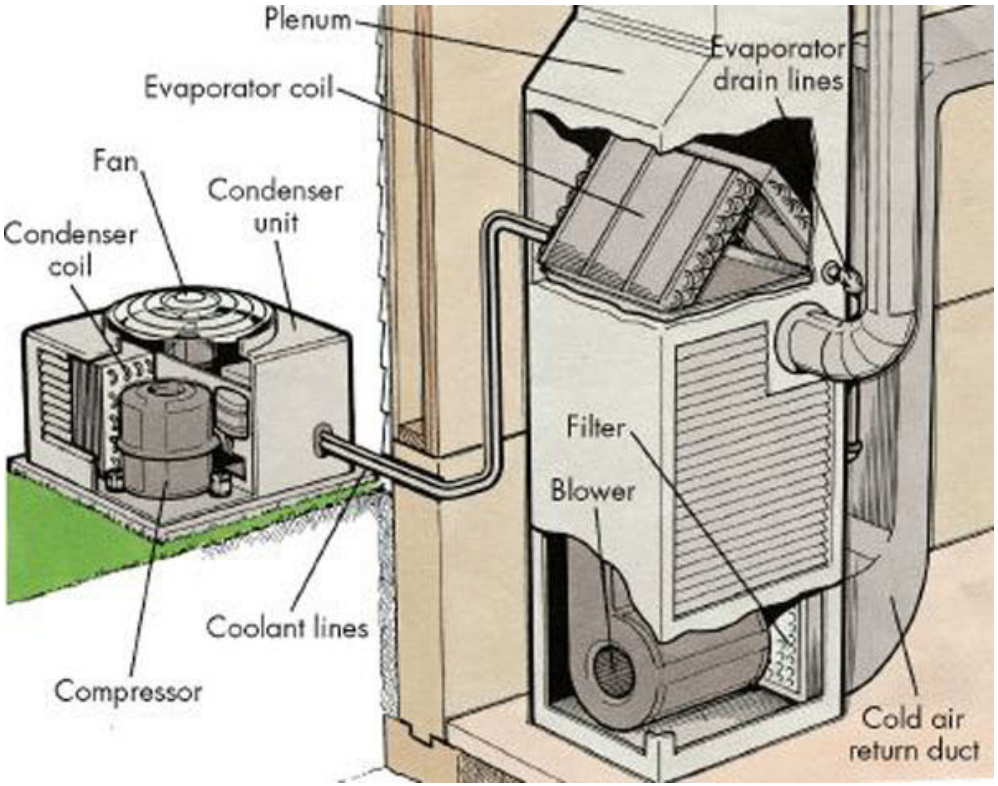




3.5 - Heating, Ventilation, and Air-Conditioning (HVAC)

3.5 – HVAC - Why Hospital HVAC Is Different

- Hospital HVAC systems are more complex than normal buildings
- Designed to support:
  - Infection prevention
  - Patient comfort
  - Regulatory compliance
- HVAC is part of the **clinical safety system**



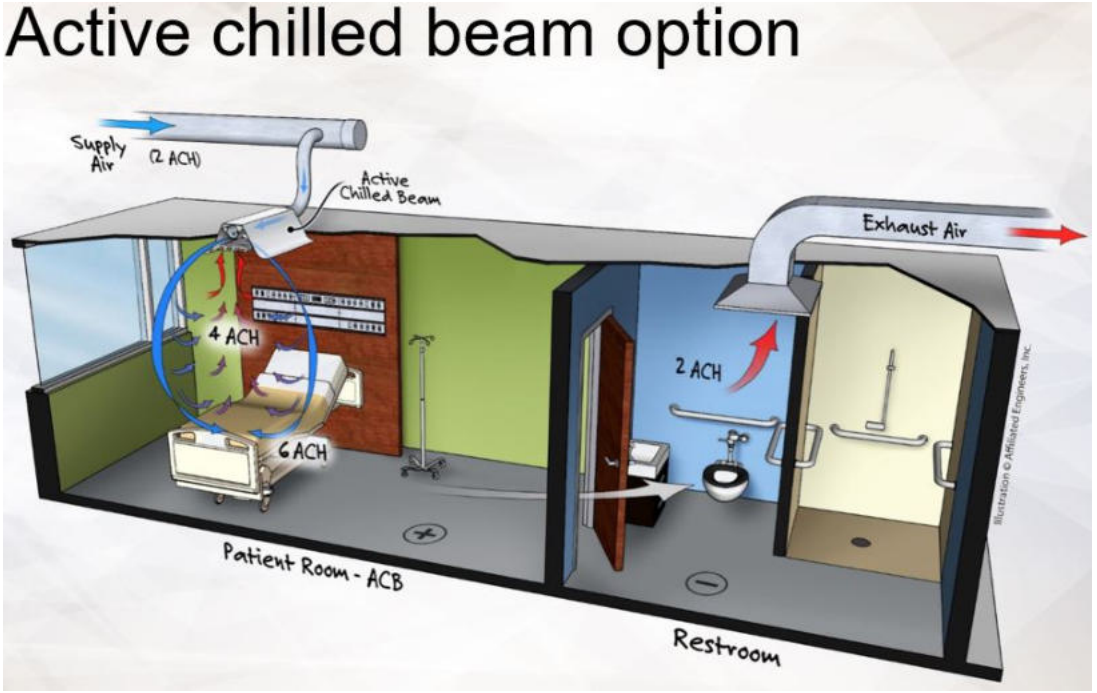
3.5 – HVAC - Strong Regulatory Standards

- Hospital HVAC follows standards from:
  - ASHRAE
  - ASHE
  - NFPA
  - ANSI
- These organizations define:
  - Airflow
  - Filtration
  - Pressure control
  - Safety requirements

3.5 – HVAC - Main Goals of Hospital HVAC Systems

- HVAC systems are designed to:
  - Control temperature and humidity
  - Remove contaminated air
  - Reduce odors
  - Protect patients from airborne infections

- Engineering goal = **safe indoor air**



3.5 – HVAC - Infection Control Through Air Handling

- HVAC helps prevent:
  - Spread of airborne pathogens
  - Cross-contamination between rooms
- Especially important for:
  - Immunocompromised patients
  - Isolation areas
  - Airflow direction is critical



3.5 – HVAC - Air Filtration as Primary Protection

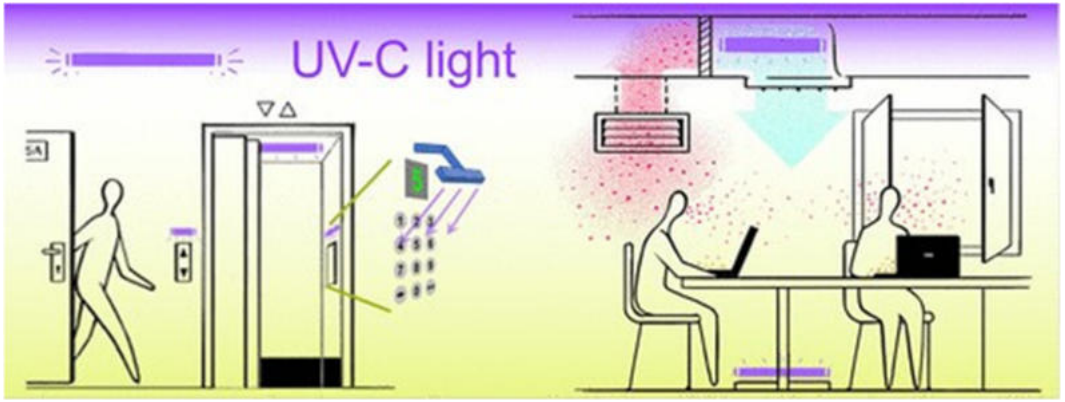
- Particles are removed using:
  - Multiple filter banks
  - Including **HEPA filters** **(**High-Efficiency Particulate Air**)**
- HEPA filters:
  - Remove **99.97%** of particles ≥ **0.3 µm**
- Used in:
  - Operating rooms
  - Isolation units
  - Clean areas



3.5 - HVAC - Supplemental Air Disinfection: UVGI

- Ultraviolet Germicidal Irradiation (UVGI):
  - Reduces bacteria and viruses in air
  - Limits fungal and bacterial growth

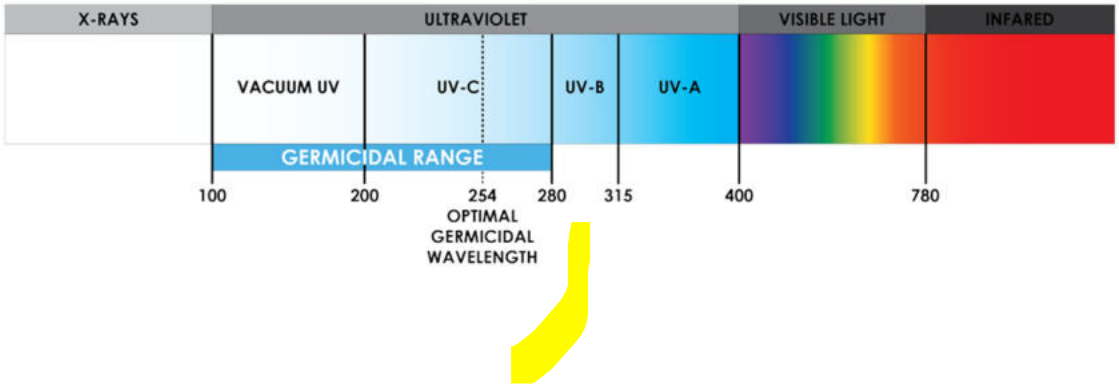
- Used as:
  - Extra protection
  - Not a replacement for filtration



3.5 – HVAC - How UVGI Works

- Most UVGI systems use:
  - Low-pressure mercury lamps
  - Emit UV at **253.7 nm**

- This wavelength:
  - Damages microbial DNA
  - Prevents reproduction



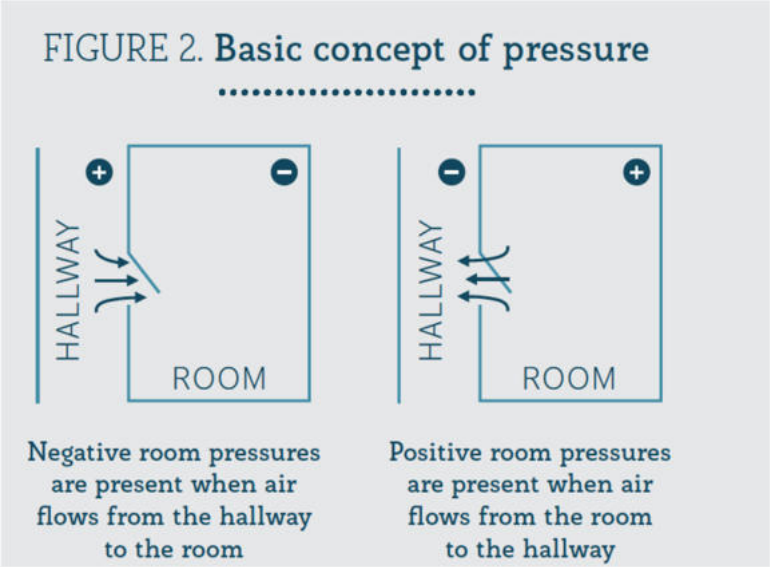
3.5 – HVAC - Limitations of UVGI

- UVGI is:
  - Effective for bacteria and viruses
  - Less effective for fungal spores
- Therefore:
- Must be combined with:
  - Filtration
  - Proper airflow design

3.6 - Positive- and Negative-Pressure Rooms

3.6 [+ & -] Pressure Rooms - What Is Room Pressurization?

- *Pressurization* = air pressure difference between:
  - **Room** and **hallway**
- Controls **direction of airflow**
- Used to control:
  - Infection spread
  - Patient protection
- Air always flows from **high pressure → low pressure**



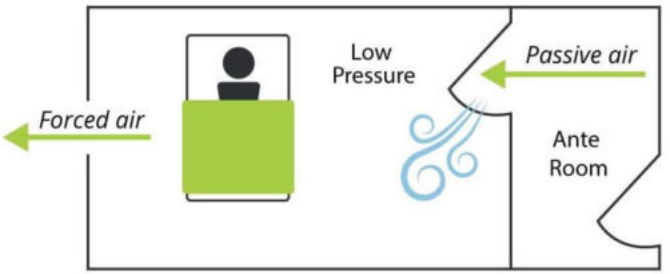
3.6 [+ & -] Pressure Rooms - Negative-Pressure Rooms (Contain Infection)

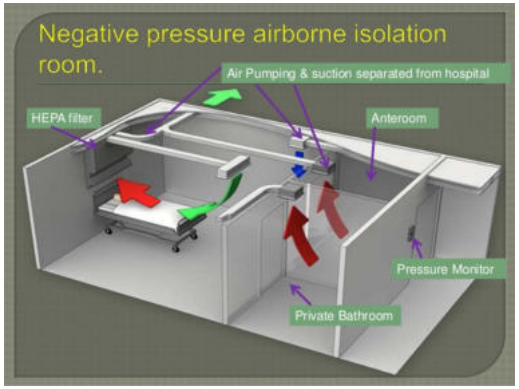
- Air flows **into the room**, not out
- Prevents contaminated air from escaping

- Used for:
  - Infectious patients
  - Contaminated procedures

- Goal: protect **others outside the room**

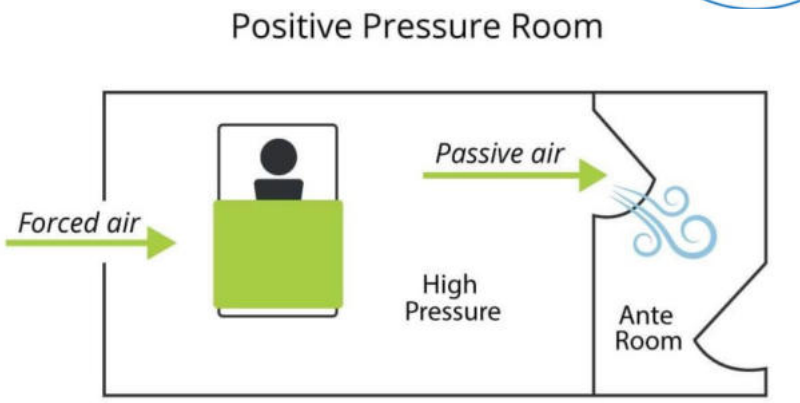
Negative Pressure Room

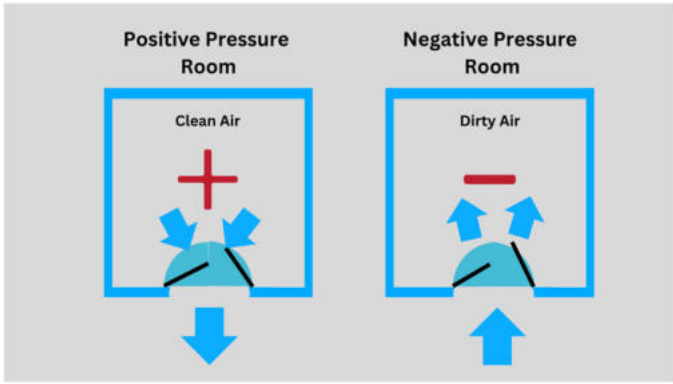




3.6 [+ & -] Pressure Rooms - Positive-Pressure Rooms (Protect Patient)

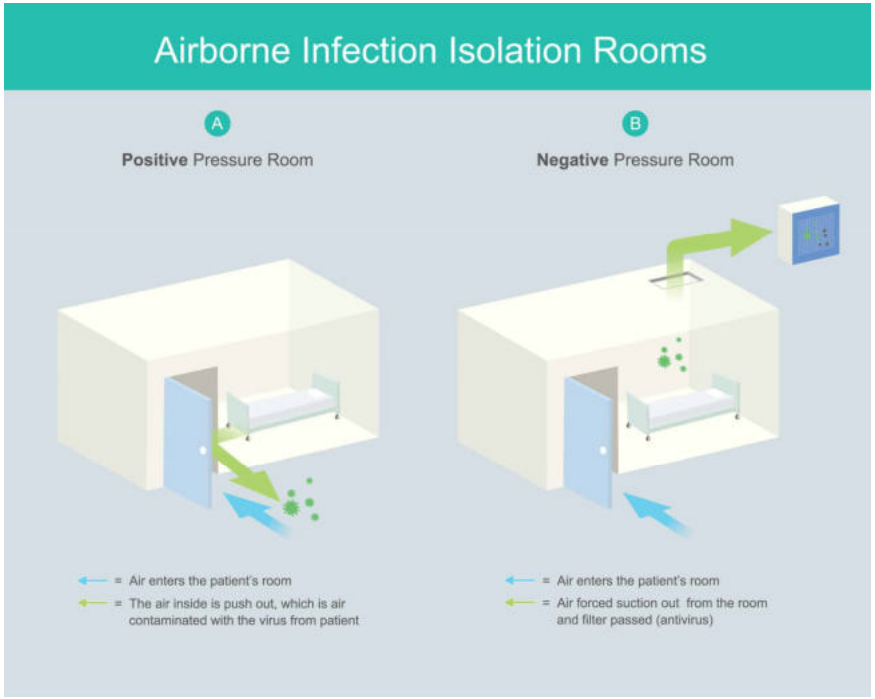
- Air flows **out of the room**
- Prevents outside air from entering
- Used for:
  - Immunocompromised patients
  - Clean procedures
- Goal: protect **patient inside the room**





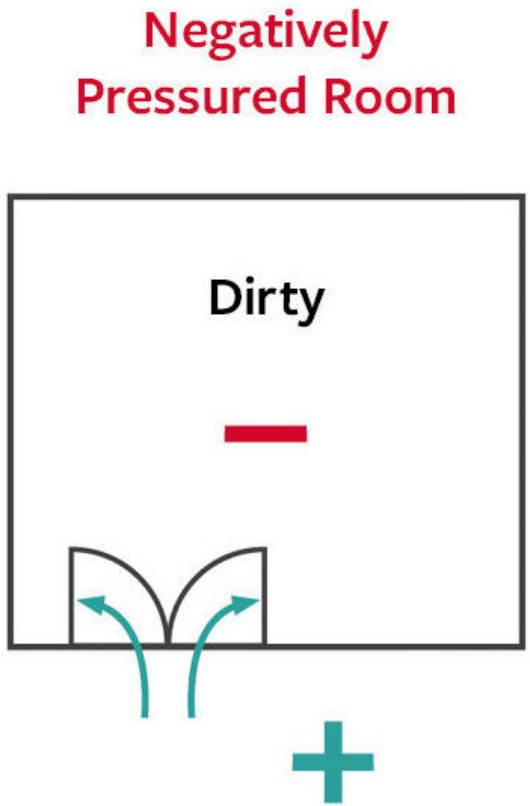
3.6 [+ & -] Pressure Rooms - Door and Structural Requirements

- Doors must be:
  - Self-closing
- Helps maintain:
  - Pressure difference
- Open doors can:
  - Break pressure control
  - Increase infection risk
- Building design supports airflow control



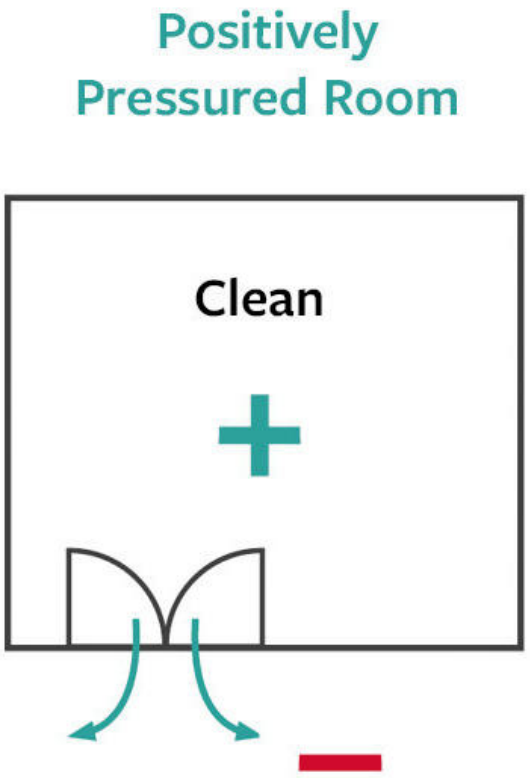
3.6 [+ & -] Pressure Rooms - Typical Negative-Pressure Areas

- Examples include:
  - Isolation rooms
  - Emergency and triage rooms
  - Pathology and microbiology labs
  - Autopsy rooms
  - Decontamination rooms
  - Soiled linen and trash rooms
  - Janitor closets and restrooms
- Used where **contamination risk is high**

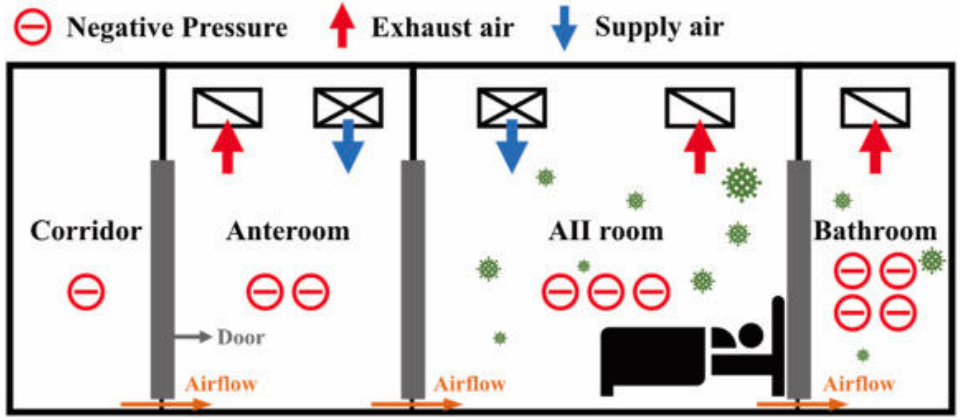


3.6 [+ & -] Pressure Rooms - Typical Positive-Pressure Areas

- Examples include:
  - Operating rooms
  - Delivery rooms
  - Trauma rooms
  - Neonatal ICUs
  - Pharmacy
  - Clean supply rooms
  - Sterile storage areas
- Used where **air must stay very clean**



3.6 [+ & -] Pressure Rooms - How Pressurization Is Controlled



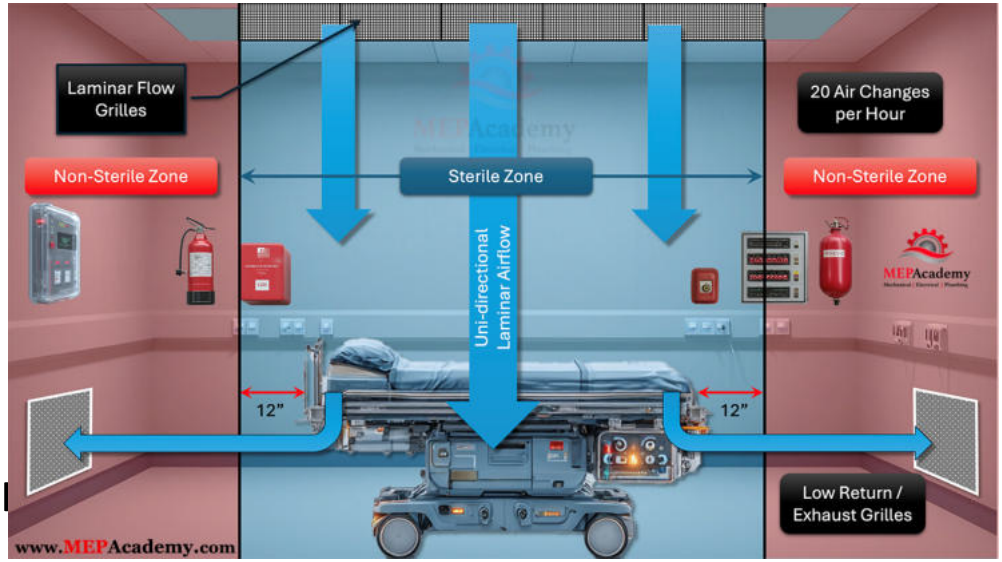
- Controlled by:
  - Supply airflow rate
  - Exhaust airflow rate
- Pressure depends on:
  - Balance between air entering and leaving room
  - Small changes can affect pressure direction



3.7 - Air-Conditioning for the Operating Room

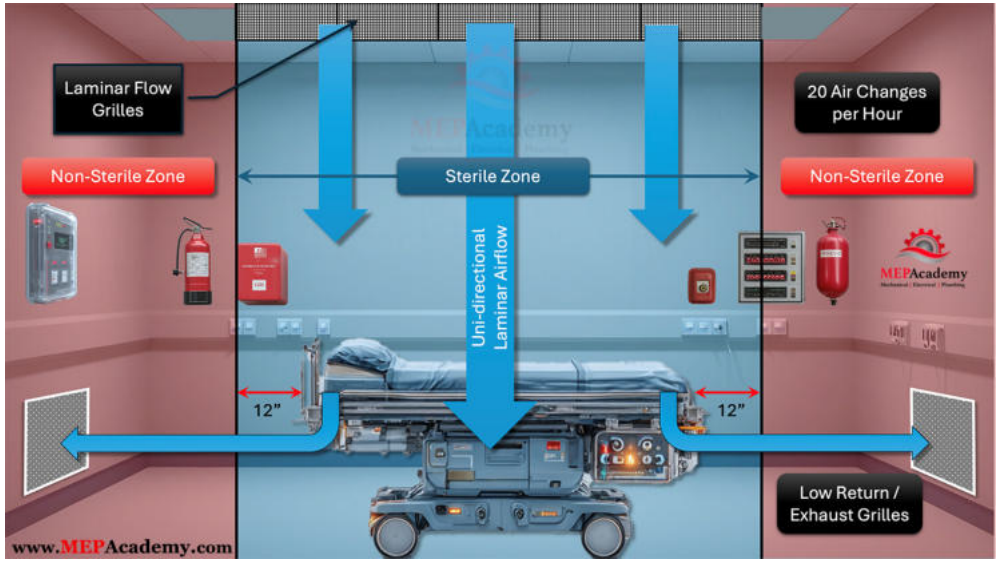
3.7 - Air-Cond.→ OR - Why OR Air-Conditioning Is Special

- Operating rooms require:
  - Very clean air
  - Controlled airflow patterns
- Purpose:
  - Reduce surgical site infections
  - Protect open surgical fields
- HVAC in OR = part of sterile barrie



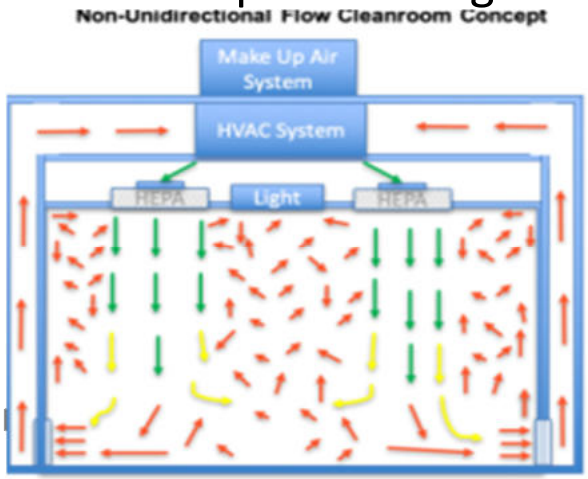
3.7 - Air-Cond.→ OR - Air Supply and Return Locations (ASHRAE 170)

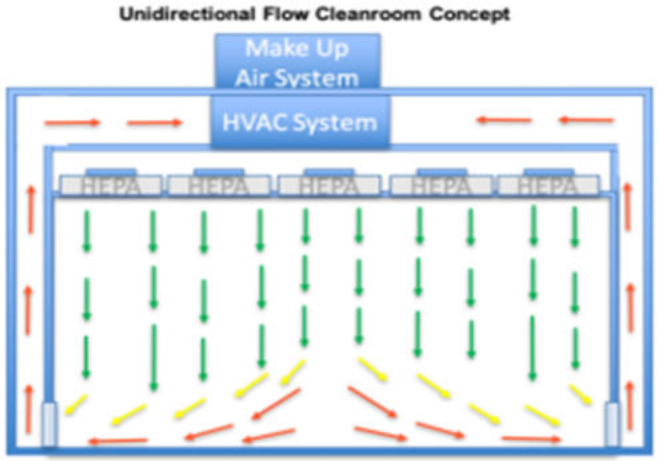
- Standards require:
  - Air supplied from the ceiling
  - Air exhausted at low wall grilles
  - At least 2 return grilles
  - Grilles located about 20 cm above floor
  - Design helps remove contaminated air downward

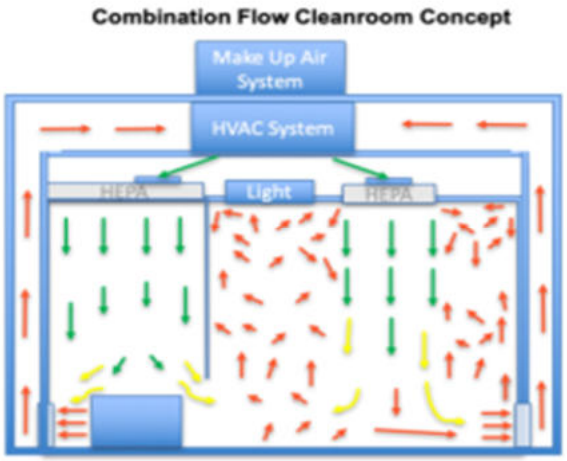


3.7 - Air-Cond.→ OR - Unidirectional (Laminar) Airflow Concept

- Air should flow in one main direction:
  - From ceiling → down to floor
- Minimizes:
  - Turbulence
  - Mixing of dirty air into sterile zone
- Goal: protect surgical area from particles

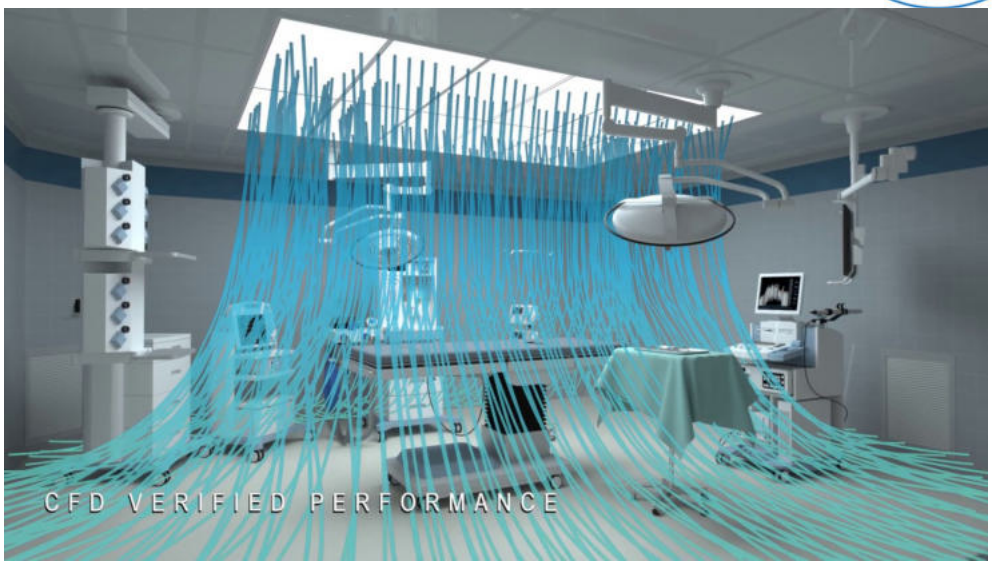






3.7 - Air-Cond.→ OR - Laminar Flow Diffusers

- Laminar diffusers:
  - Create smooth, uniform airflow
  - Reduce random air currents
- Installed above:
  - Operating table
  - Surgical field



- Other equipment areas may receive non-laminar air

3.7 - Air-Cond.→ OR - Air changes per hour

- Standard requirement:
  - **20 *air changes per hour* (ACH)** in OR
  - The air inside the room is completely replaced **20 times in one hour**
  - The room gets fresh air about **every 3 minutes**
- Purpose:
  - Remove contaminants quickly
  - Maintain air cleanliness
  - Higher ACH = faster air replacement

3.7 - Air-Cond.→ OR - Higher ACH in Special Surgeries

- Historically ,**40 ACH** used in:
  - Orthopedic surgery
  - Open-heart surgery
- But:
  - No strong evidence that higher ACH always reduces infection
- So:
  - 20 ACH is standard practice

3.8 - Anesthetic Gases

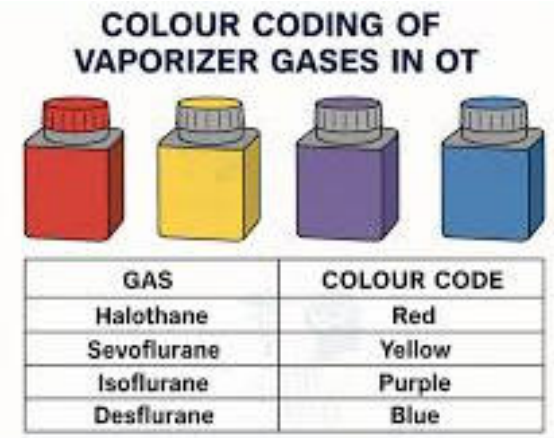
3.8 - Anesthetic Gases - Anesthetic Gases in Hospitals

- Anesthetic gases are inhaled medications used to:
  - Induce unconsciousness
  - Control pain
  - Relax muscles during surgery
- Used in:
  - Operating rooms
  - Procedure rooms
  - Emergency surgery



3.8 - Anesthetic Gases - Common Anesthetic Gases Used

- **Most Common Gases:**
  - Nitrous Oxide (N_2O) – very widely used
  - Sevoflurane
  - Isoflurane
  - Desflurane



GAS	COLOUR CODE
Halothane	Red
Sevoflurane	Yellow
Isoflurane	Purple
Desflurane	Blue

- N_2O is historically one of the most common anesthetic gases, especially for short procedures and dentistry.
- Modern major surgeries often rely more on sevoflurane and similar agents.

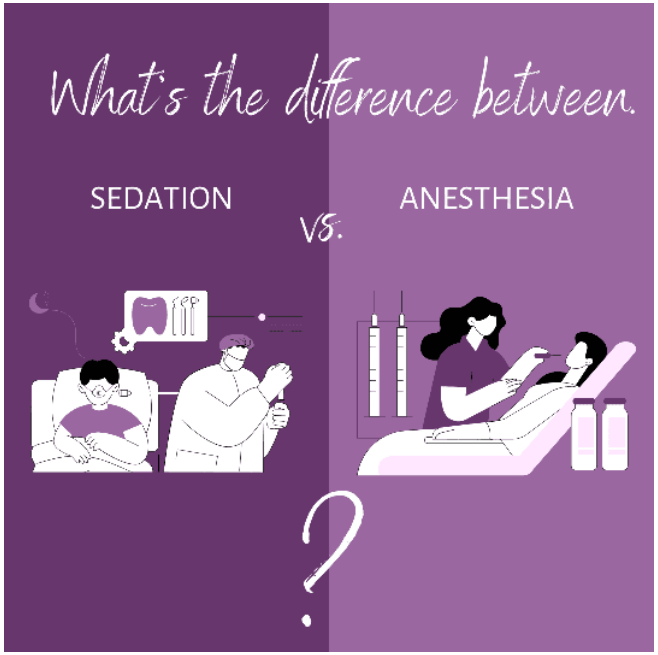
3.8 - Anesthetic Gases - Engineering Requirements

- Anesthetic gases require:
  - Central storage (cylinders or tanks)
  - Medical-grade piping
  - Pressure regulation
  - Alarm and monitoring systems
- Failure can affect:
  - Patient safety
  - Surgical procedures



3.8 - Anesthetic Gases - Anesthesia and Sedation in Hospitals

- **Anesthesia :** A controlled medical state that causes:
  - Loss of consciousness
  - Loss of pain sensation
  - Muscle relaxation
  - Used for major surgeries
- **Sedation:** A reduced level of consciousness that:
  - Relaxes the patient
  - Reduces anxiety
  - May relieve mild pain
  - Used for: Minor procedures ,Endoscopy, Dentistry



4 - Building and renovation project management

4 - Building and renovation project management - Why Clinical Engineering Is Involved

- Clinical engineering supports:
  - Renovations
  - New hospital construction
- Main role:
  - Plan and manage **medical equipment needs**
- Goal:
  - Ensure rooms are ready for safe equipment use

4 - Building and renovation project management - Equipment Planning Responsibilities

- Depending on project size:
- **Small projects**
  - Clinical engineers:
    - Select equipment
    - Get quotes
    - Order
    - Support installation
- **Large projects**
  - Third-party equipment planners may be used

4 - Building and renovation project management - Architecturally Significant Equipment

- Equipment that directly affects building design
- Must be planned early in the project
- **Typical Requirements**
  - Power • Water • HVAC • Data
- **Special Considerations**
  - Very heavy → **structural reinforcement**
  - May require **shielding** (radiation / magnetic fields)
- **Examples**
- MRI / CT scanners / PET scanners / LINACs … etc.
- Pneumatic tube systems
- Ceiling-mounted booms



5 - Emergency preparedness

5 – Emergency - Why Emergency Preparedness Is Essential

- Hospitals must stay operational during:
  - Disasters
  - Accidents
  - System failures
- Patient care cannot stop during emergencies
- Preparedness = **continuity of care**

5 – Emergency - Hospitals Must Plan for Many Event Types

- Emergency plans may include:
  - Weather-related events
  - Security incidents
  - Large-scale disasters
  - Cyberattacks on hospital systems
- Each type needs:
  - Different response strategies

5 – Emergency - Importance of Drills and Training

- Departments regularly:
  - Practice emergency scenarios
  - Test communication and response

- Drills help identify:
  - Weak points
  - System failures before real events



5.1 - Weather-Related Events

5.1 - Weather-Related Events - What Are Weather-Related Emergencies?

- Examples include:
  - Earthquakes
  - Floods
  - Wildfires
  - Tornados and hurricanes
  - Extreme heat or winter storms
  - Landslides and tsunamis
- These events can:
  - Damage infrastructure
  - Disrupt patient care

5.1 - Weather-Related Events - Risk Depends on Hospital Location

- Not all hospitals face the same risks:
- Coastal → hurricanes, floods
- Seismic zones → earthquakes
- Hot climates → extreme heat
- Cold regions → snow and ice
- Emergency plans must match **local risks**

5.1 - Weather-Related Events - One Plan Does NOT Fit All

- Different disasters cause different problems:
  - Power failure
  - Access blocked
  - Staff unable to reach hospital
  - Water or gas interruptions

- Therefore:
  - Separate response plans are needed for each event type

5.1 - Weather-Related Events - What Emergency Plans Must Include

- Each plan should define:
  - Who responds (staff and leadership)
  - What supplies are needed
  - Where resources should go
  - How to communicate
  - Clear roles reduce confusion during crisis



5.2 - Cyberattacks

5.2 – Cyberattacks - Why Cyber Preparedness Matters

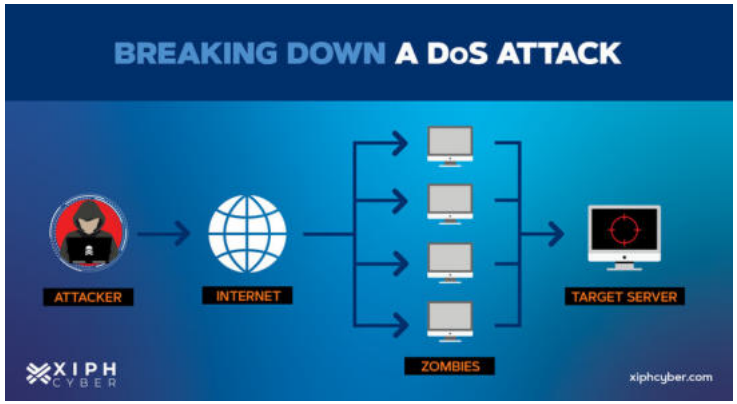
- Hospitals depend on:
  - Computer systems
  - Networks
  - Electronic patient records
- Cyberattacks can:
  - Disrupt care
  - Delay treatments
- Emergency planning must include **IT failures**

5.2 – Cyberattacks - Main Types of Cyberattacks

- Common hospital cyber threats:
  - **Ransomware attacks**
  - **Denial-of-Service (DoS) attacks**



- These attacks mainly target:
  - Hospital IT systems
  - Patient data systems



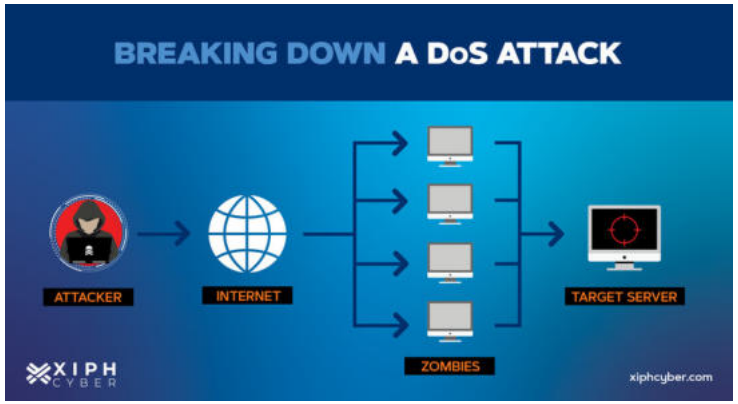
5.2 – Cyberattacks - What Is Ransomware?

- Malicious software that:
  - Locks computer systems
  - Blocks access to files
- Access is restored only after:
  - Payment is demanded
- Often spreads through:
  - Phishing emails
  - Infected websites



5.2 – Cyberattacks - What Is a DoS Attack?

- Overloads a network or server
- Makes systems:
  - Slow
  - Unusable
- Staff cannot access:
  - Patient records
  - Scheduling systems
  - Communication tools
- Care delivery may be delayed



5.2 – Cyberattacks - Who Is Responsible for Cyber Protection?

- Primary responsibility:
  - IT departments

- But clinical engineers must also be aware because:
  - Medical devices are increasingly networked
  - Device connectivity creates new risks

- Cybersecurity is now **multidisciplinary**

5.2 – Cyberattacks - Staff Behavior Is Also Critical



- All hospital staff must follow:
  - Safe internet use
  - Strong passwords
  - Phishing awareness
- Many attacks start with: Human error

- Training reduces risk significantly

6 - Infection prevention

6 - Infection prevention - What Is Infection Prevention?

- Prevents spread of infectious diseases inside hospitals
- Protects:
  - Patients
  - Staff
  - Visitors
- Part of overall **patient safety system**



6 - Infection prevention - Why Infection Prevention Is Critical

- About **1 in 25 hospitalized patients** gets a hospital-acquired infection

- These infections can:
  - Delay recovery
  - Increase hospital stay
  - Increase risk of death

- Prevention saves lives and resources



6 - Infection prevention - Key Areas of Monitoring



- Hospitals track common hospital-acquired infections such as:
  - Urinary tract infections from catheters
  - Surgical site infections
  - Bloodstream infections from IV lines
  - Drug-resistant bacteria infections
  - Pressure-related skin injuries
  - Tracking helps identify problem areas

6 - Infection prevention - Hand Hygiene Is a Major Focus

- Hand hygiene is the **most effective prevention method**
- Programs aim to:
  - Improve staff compliance
  - Reduce cross-contamination
- Simple actions have big impact



6.1 - Hand hygiene

6.1 - Hand hygiene - What Is Hand Hygiene?

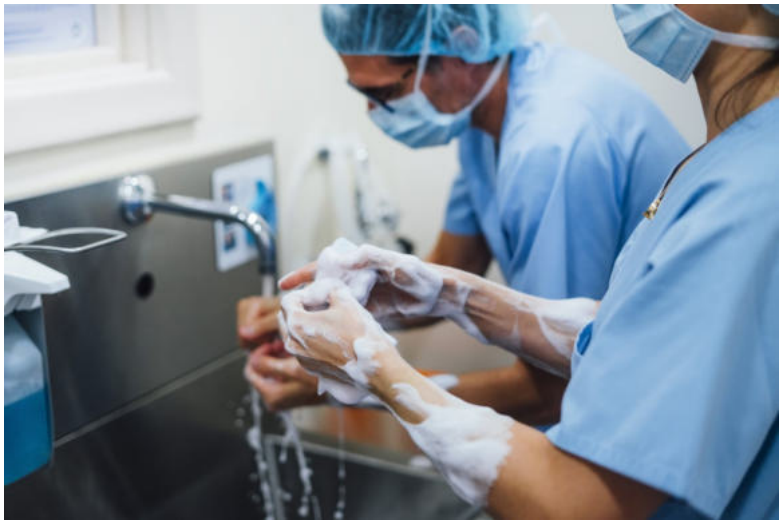
- Hand hygiene means cleaning hands using:
  - Soap and water
  - Alcohol-based hand sanitizer
  - Surgical hand antisepsis (before surgery)
- Goal:
  - Remove germs
  - Prevent infection spread





6.1 - Hand hygiene - Why Hand Hygiene Is So Important

- Hands are the main way germs spread in hospitals
- Proper hand hygiene:
  - Protects patients
  - Protects staff
- Most effective infection-prevention action



6.1 - Hand hygiene - Most Effective Method

- When hands are **not visibly dirty**:
  - Alcohol-based hand sanitizer is best
  - Fast
  - Kills most germs effectively
- Used in:
  - Patient rooms
  - Hallways
  - Bedside care



6.1 - Hand hygiene - Second Best Option

- Antiseptic soaps are also effective
- Used when sanitizer is not available
- Requires:
  - Sink
  - Running water



6.1 - Hand hygiene - When to Use Soap and Water (Not Sanitizer)

- Soap and water are required when:
  - Hands are visibly dirty
  - Before eating
  - After using restroom

- After contact with patients with:
  - Diarrhea
  - Suspected infection outbreaks

6.1 - Hand hygiene

- **Special Cases: Certain Infections**
- Alcohol sanitizer is **not enough** after exposure to:
  - **C. difficile**
  - **Norovirus**
- These require:
  - Soap and water washing
  - Because spores are not easily killed by alcohol

6.1 - Hand hygiene - Surgical Hand Antisepsis



- Used before:
  - Surgical procedures
- Purpose:
  - Remove normal skin bacteria
  - Prevent surgical site infections
  - More thorough than routine hand cleaning





Think with me !? 🤔



- In one line , What is the difference between sedation and general anesthesia?



Thank you